Music Player – Python Mini Project Report

By
Tom Ison
S6 CSE-C
Roll no: 63

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Abstract

This project presents a desktop music player application developed in Python, leveraging libraries beyond the standard syllabus. The player offers a modern interface, playlist management, lyrics display, and advanced playback controls. It demonstrates practical use of Python for multimedia applications, integrating third-party libraries for audio processing, metadata extraction, and GUI enhancements. The solution is stable, user-friendly, and extensible for future features.

Problem Description

The goal is to create a modern music player for desktop users, capable of playing audio files, displaying metadata and lyrics, and providing a visually appealing interface. The project demonstrates how Python can be used for multimedia applications, integrating advanced libraries for enhanced functionality.

Libraries & Tools Used

- **mutagen**: Extracts audio metadata (title, artist, album, lyrics, album art).
- **vlc**: Handles audio playback and advanced controls.
- **Pillow (PIL)**: Processes album art images.
- **customtkinter**: Provides a modern, customizable GUI.
- **pypresence**: Integrates Discord Rich Presence (optional).

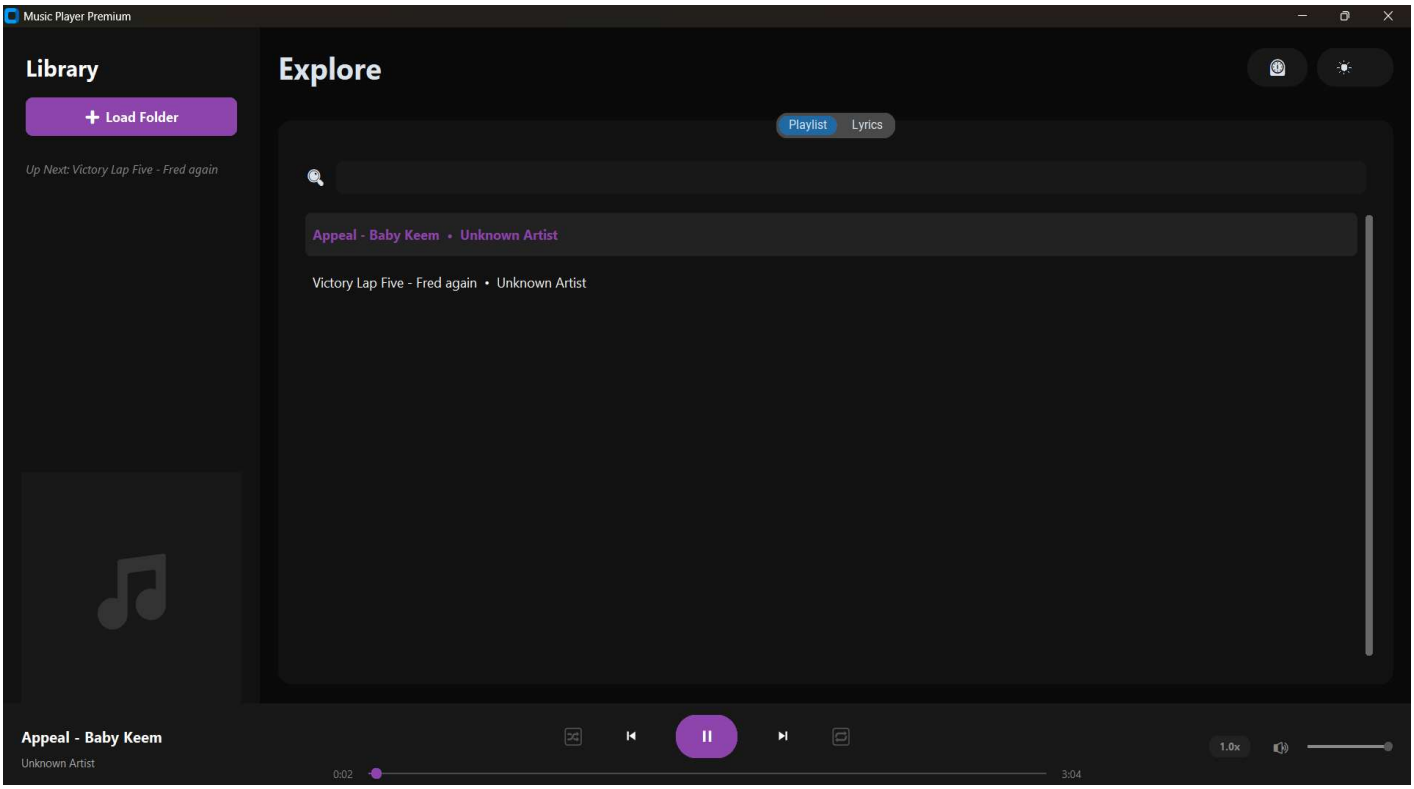
System Design/Workflow

- The user loads a folder containing music files.
- The app reads metadata and displays songs in a playlist.
- Lyrics and album art are shown if available.
- Playback controls allow play, pause, shuffle, repeat, speed, and mute.
- The interface supports light/dark themes.

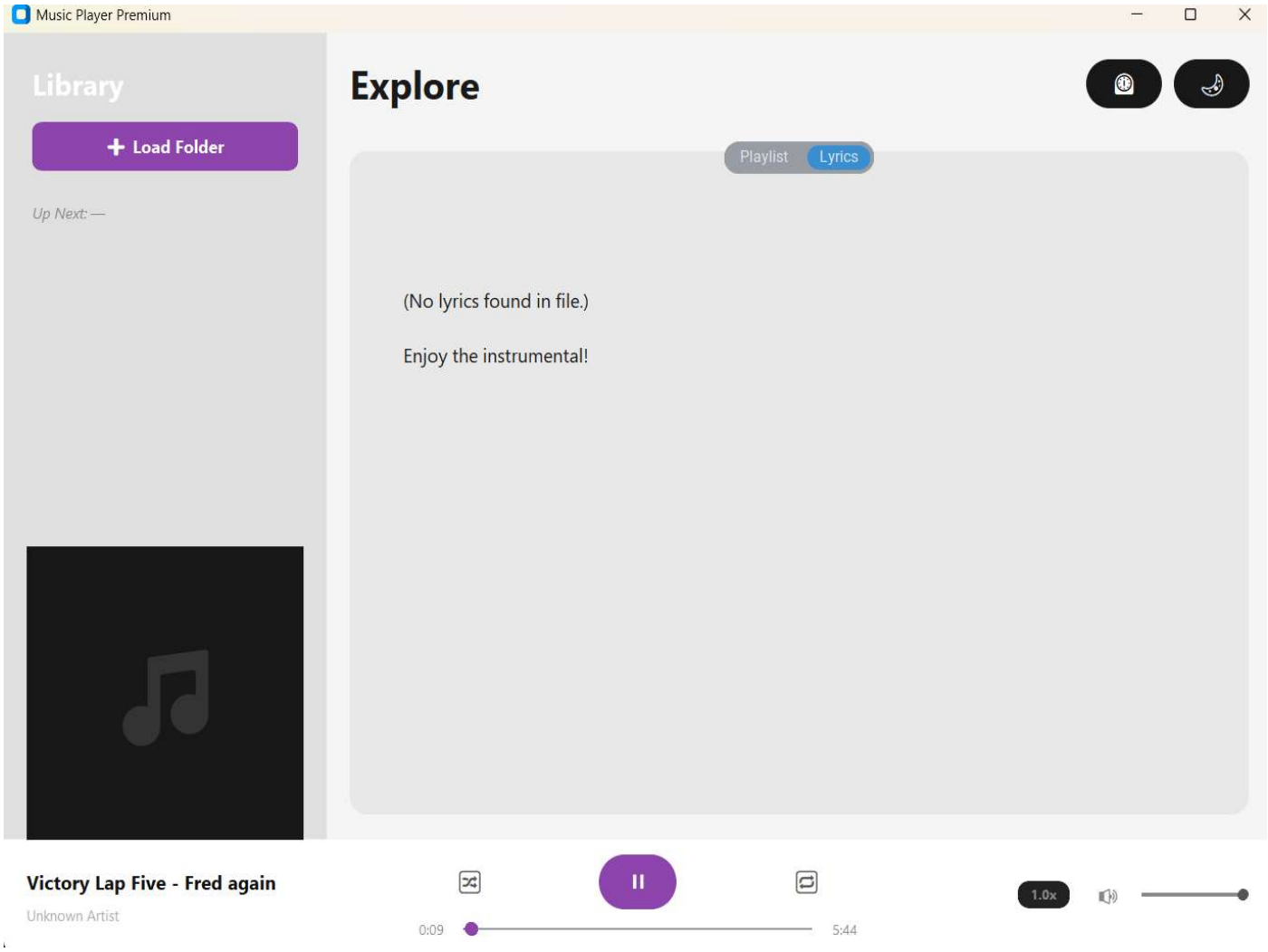
Implementation Details

The application is built using customtkinter for the GUI, with vlc for playback. Metadata is extracted using mutagen, and album art is processed with Pillow. The app supports playlist management, lyrics display, and advanced playback features.

Results & Output



Starting page (Dark Mode)



(Light Mode)

Conclusion & Scope

The music player is fully functional and demonstrates the use of advanced Python libraries for multimedia applications. Future extensions could include online streaming, playlist export, equalizer, and more advanced lyrics fetching.

References

- mutagen documentation
- vlc Python bindings
- Pillow documentation
- customtkinter documentation